

Skills & Connectors

Hands-On Project Guide — 5 Projects, with Diagrams

Source: *The AI Agent Factory* — Skills & Connectors: Teach AI Once, Connect It to Your Apps (agentfactory.panaversity.org). Do the projects in order — Project 3 needs 1 and 2 finished first. Each one fits a free Claude.ai account.

Before You Start

1. Open Claude.ai in a browser or the app (a free account is enough).
2. Turn on file creation: go to Settings, then Capabilities, and switch on **Code execution and file creation**. This powers the built-in skills you'll use.
3. For Projects 2 and 3 you'll connect a real app — Google Drive or Gmail is the easiest. Have one of those accounts ready.
4. Keep a real (or realistic) piece of your own work handy — a set of class notes, a report, or a document in your Drive — so you have something genuine to test on.

What's in this guide

Project 1	Build your first real skill	30–45 min
Project 2	Connect one app, read-only, and use it	20–30 min
Project 3	Wire a skill and a connector together	45–60 min
Project 4	Make it portable, or hand it off	30 min
Project 5	Audit a skill before you trust it	15–20 min

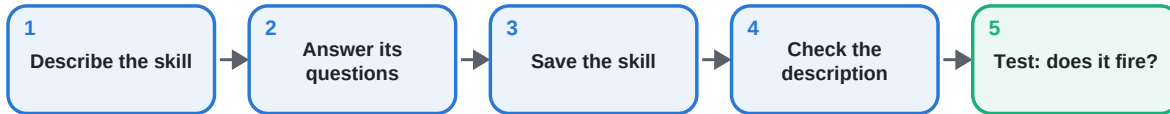
A quick way to test yourself as you go: for each project, before you start, answer — is the friction here re-explaining *how* (a skill), fetching data from an app (a connector), or both? That's the same test the book uses in section 9.

PROJECT 1 • Build your first real skill

Time: 30-45 min

Turn your most re-explained task into a skill that just knows your way.

At a glance



How to do it

1. Open a new chat in Claude.ai.
2. Paste the prompt below, word for word, to build a real skill by talking to it.

Paste this, word for word:

```
Use the skill-creator skill to build me a weekly study-notes skill. Whenever I ask for "study notes" or "weekly notes", take my raw class notes and format them as a heading per topic, a short bold key-terms list, and exactly three review questions at the end. Ask me anything you need, then build it.
```

1. Answer skill-creator's clarifying questions (it will ask a few, e.g. exactly how you want things formatted).
2. When it's done, click **Save skill**. It lands on your shelf at Customize, then Skills — private to your account, toggled on.
3. Check the description (the part that decides whether the skill fires on its own). Paste:

Paste this, word for word:

```
When would you use the weekly-study-notes skill, and when would you NOT use it?
```

1. If the answer is too wide or too narrow, tell it the fix in plain English.
2. Now prove it works. Open a **brand-new chat**, and — without naming the skill — paste a real week of your notes right after this line:

Paste this, word for word:

```
Make study notes from this:
```

Watch it reach for your skill on its own and hand back your exact format every time: a heading per topic, key terms in bold, three review questions.

To make this about your own class-prep task instead of study notes: keep the exact same three-message shape above, and swap in the thing you keep re-typing for your students — your MCQ format, your slide-deck outline rules, your grading rubric — with your own must-dos and must-nots.

How to test it / done when:

- The skill fires on its own from a natural request — no slash command, no naming it.
- It produces your exact format without you reminding it.
- It does NOT take over an unrelated request.

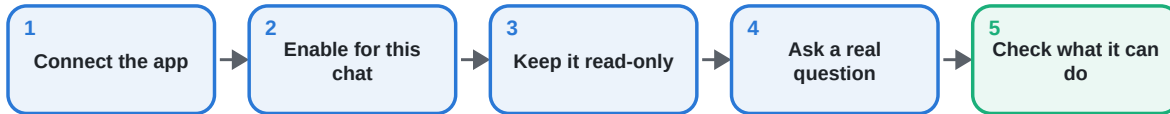
The win you can say out loud: "I taught AI how I do my weekly notes once, and now it does it in one sentence, every time."

PROJECT 2 • Connect one app, read-only, and use it

Time: 20-30 min

Get a real answer straight from your own data, with zero copy-paste.

At a glance



How to do it

1. Click the "+" in the chat box (or open Customize, then Connectors), and connect Google Drive or Gmail — whichever you actually use.
2. Complete the sign-in / authorization prompt, the same "allow this app to access your account" flow you already know.
3. Enable the connector for this conversation (toggle it on from the same "+" menu).
4. Keep it **read-only** for now — don't grant write/send/delete access yet.
5. Ask a real question that needs something inside the app. For Drive, paste:

Paste this, word for word:

Find my most recent document in my Drive, pull the three facts or numbers that matter most, and give me a one-line summary of what it is.

Or for Gmail:

Paste this, word for word:

Summarize my most recent email thread, and list anything it is waiting on me to do.

1. Confirm you never uploaded or pasted anything — the answer came straight out of your own account.
2. Say out loud exactly what access you granted, and decide: will this task ever need write access? If so, what would you want to see before allowing it?

How to test it / done when:

- You pulled real data through a connector, not by copy-pasting.
- You can state in one sentence what the connector can and cannot do with your account.

The win you can say out loud: "I asked one question and AI pulled the answer out of my own files; I never copied or pasted a thing."

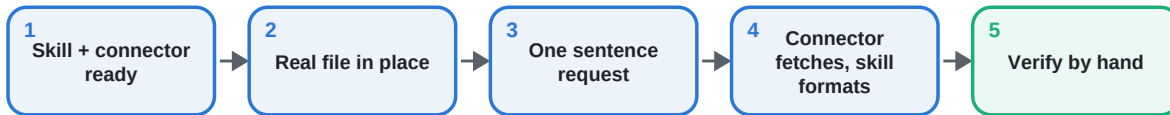
PROJECT 3 • Wire a skill and a connector together

Time: 45-60 min

The monthly-close / weekly-batch machine, running on your own data.

Needs Projects 1 and 2 done first.

At a glance



How to do it

1. Confirm your Project-1 skill is on your shelf and toggled on, and your Project-2 connector is enabled for this chat.
2. Put a real file where the connector can reach it — for example, this week's class notes saved in your Drive.
3. Send one plain sentence that needs both the connector and the skill:

Paste this, word for word:

Find this week's class notes in my Drive and make study notes from them.

For a report-style version (the accountant's shape from the book, same idea, different data):

Paste this, word for word:

Prepare this month's summary for DEMO Trading; the ledger is the file called DEMO-ledger-March in my Drive.

1. Watch the connector fetch the file and the skill format it — you just review the result.
2. Verify the output against one part you can check by hand before you trust the whole thing.

How to test it / done when:

- One plain-English sentence produces a finished, correctly-formatted result.
- The result is built from live data you never copied or pasted yourself.

The win you can say out loud: "I taught AI my weekly report once and pointed it at my files; now the whole thing runs from one sentence and I just review it."

PROJECT 4 • Make it portable, or hand it off

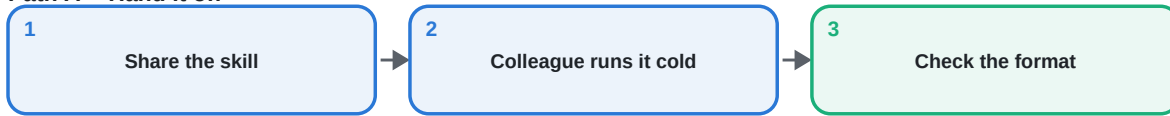
Time: 30 min - capstone

Prove the asset outlives the chat AND the tool.

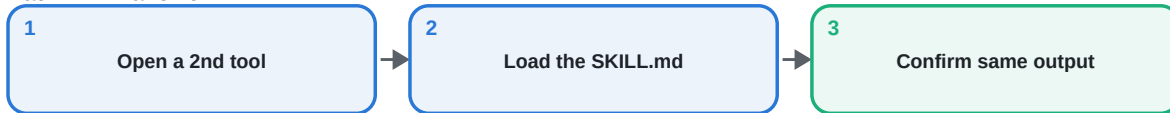
Pick ONE of the two paths below.

At a glance

Path A -- Hand it off



Path B -- Travel it



Path A: Hand it off

1. Share your Project-1 skill with a friend or colleague (or publish it to your organization's directory, if you're on a Team or Enterprise plan).
2. Have them run it completely cold — nothing but the skill itself, no explanation from you.
3. Check whether they get your exact format without any help.

Path B: Travel it to a second tool

1. Open a second surface — Cowork or Claude Code.
2. Load the same SKILL.md file there.
3. Paste a real week of notes right after this line:

Paste this, word for word:

Use this skill to make study notes from:

Confirm it produces the same correct output in the new tool.

How to test it / done when:

- Someone who was not in the original conversation, OR a second tool, produces a correct result from your skill.
- This happens with no help from you in the moment.

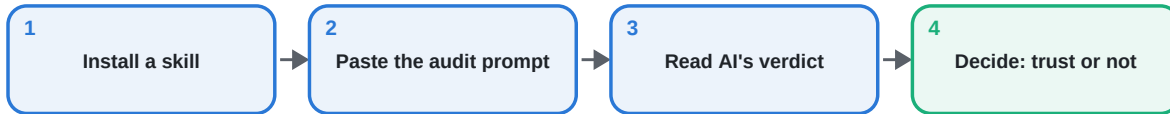
The win you can say out loud: "I handed my skill to a friend and it just worked, without me in the room."

PROJECT 5 • Audit a skill before you trust it

Time: 15-20 min

Install one skill from a stranger, then prove it's safe before you let it run.

At a glance



How to do it

1. Open the official directory: Customize, then Skills, then "+", then Browse skills.
2. Install one skill you did NOT write yourself.
3. Before you rely on it for anything real, paste this audit prompt:

Paste this, word for word:

```
Read the skill I just installed and tell me, in plain language, exactly what it instructs you to do. Then flag anything in it that contacts an external server, handles passwords or credentials, or could send my data somewhere I didn't intend. If it's clean, say so plainly.
```

1. Read the plain-language explanation and the verdict: clean, or "here is the line that reaches the internet."
2. Decide whether to keep it enabled based on that verdict — not on how official it looked in the directory.

How to test it / done when:

- You can say in one sentence what a skill you didn't write actually does.
- You know whether it reaches anything outside your own account.

The win you can say out loud: "I read a skill before trusting it, in plain English, without understanding a single line of code."

If You Get Stuck

Skill never fires, or fires on the wrong things — Description problem. Ask: "When would you use my [name] skill, and when would you not?" Then tighten the description — add the exact words you say, or a "Do NOT use for..." line.

Connector "can't find" a file — Usually permissions or scope. Confirm the connector is enabled for this chat, and that your own account can actually open that file.

AI answers from a guess instead of the connected data — Say it explicitly: "Use my [app] connector to fetch the actual file before answering, and tell me which file you used."

Nervous about write access or sensitive data — Stay read-only. Grant write access only to a tool that has already behaved well, and only for the smallest folder the task needs.

Chat has gotten long and confused — Start a fresh chat and restate the task in two lines. A tangled chat is cheaper to abandon than to rescue.

A skill is a set of instructions you're letting AI follow. A connector is a door into your real data. Read a skill before you enable it, and start every connector read-only.